

Vitamins Co-Enzymes, Functions, and Symptoms Related to Deficiency or Toxicity



Vitamin B-Complex

	B1/Thiamin (mg·d ⁻¹)	B2/Riboflavin (mg·d ⁻¹)	B3/Niacin (mg·d ⁻¹)	B5/Pantothenic Acid (mg·d ⁻¹)	B6/Pyridoxine (mg·d ⁻¹)	B7/Biotin (µg·d ⁻¹)	B9/Folic Acid (µg·d ⁻¹)	B12/ Cobalamin (µg·d ⁻¹)
Forms Major Co-Enzymes	Thiamin diphosphate (TDP)	Flavin adenine dinucleotide (FAD); flavin mononucleotide (FMN)	Nicotinamide adenine dinucleotide (NAD); nicotinamide adenine dinucleotide phosphate (NADP)	Coenzyme A (CoA) and 4' phosphopantetheine	Pyridoxal phosphate (PLP)	Carboxybiotin	Tetrahydrofolate (THF) derivatives: 5,10-methylene THF, 10-formyl THF, 5-formimino THF, 5,10-methylenyl THF, 5-methyl THF	Methylcobalamin and adenosylcobalamin
Cellular or Physiological Role	Promotes CHO metabolism and CNS function	Electron (hydrogen) transfer reactions for nutrient metabolism and energy production (CHO and fat oxidation) and supports healthy skin	Electron (hydrogen) transfer reactions for nutrient metabolism and energy production (CHO and fat oxidation), and fat synthesis; supports healthy skin	Promotes CHO and fat oxidation, lipogenesis, and gene expression	Promotes protein metabolism, formation of hemoglobin and RBCs, glycogenolysis, gluconeogenesis, and gene expression	Promotes CHO, fat, and protein metabolism, gene expression	Promotes formation of hemoglobin, RBCs, WBCs, and gene expression	Promotes nutrient metabolism, energy production, and formation of RBCs and WBCs; maintains nerve, gut, and skin tissue
Deficiency	Beriberi -Nervous system and cardiac dysfunction; apathy, depression	Edema oral cavity, dermatitis, lip and tongue sores, damage to the cornea of the eye	Pellagra diarrhea, dermatitis, mental confusion, weakness; skin lesions, gut and skin issues	Nausea, fatigue, depression, loss of appetite, burning of the feet, neuritis	Irritability, dermatitis, convulsions, anemia, nervous system dysfunction, and tongue sores	Dermatitis, hypotonia, nervous system dysfunction, alopecia, nausea, fatigue, skin rashes	Megaloblastic anemia (enlarged immature red blood cells), diarrhea, fatigue, gut disorders, and infections	Megaloblastic anemia, degeneration of peripheral nerves, paralysis, infections
Excess or Toxicity	Not reported	Not reported	Headache, nausea, skin irritations, liver damage, inhibition of lipolysis	Not reported	Loss of nerve sensation, abnormal gait	Not reported	Not reported	Not reported

Carbohydrate (CHO); central nervous system (CNS); red blood cells (RBCs); white blood cells (WBCs). Adapted from Gropper & Smith (2022) and Jeukendrup & Gleeson (2018).

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Vitamin

	Vitamin C	Choline
Forms Major Co-Enzyme(s)	None	None
Cellular or Physiological Role	Antioxidant; promotes collagen formation, development of connective tissue, catecholamine and steroid synthesis, and iron absorption	Required for proper liver, muscle, and brain functions; lipid metabolism; and cellular membrane composition and repair
Deficiency	Scurvy, hyperkeratosis of hair follicles, slow wound healing, impaired collagen synthesis, bleeding gums, anemia, and ruptured capillaries	Choline-deficient diet develops a fatty liver disease, elevated liver enzymes in the blood, muscle damage, an increase in creatine kinase concentrations; UL of 3.5 g·d ⁻¹ for adults
Excess or Toxicity	No toxic effects when consumed in smaller doses (<1,000 mg·d ⁻¹); diarrhea, kidney stones, and excess iron absorption	Fishy body odor, vomiting, excessive sweating and salivation, hypotension, and liver toxicity

Tolerable Upper Intake Level (UL). Adapted from Beck et al. (2021), Gropper & Smith (2022) and Wallace et al. (2018).

Fat-Soluble Vitamins

	Vitamin A/Retinol, Retinal, and Retinoic Acid, Carotenoids	Vitamin D/D ₂ - Ergocalciferol, D ₃ - Cholecalciferol, Provitamin - 7-Dehydrocholesterol	Vitamin E/Tocopherols and Tocotrienols	Vitamin K/Phylloquinones and Menaquinones
Cellular or Physiological Role	Maintains epithelial tissues of the skin, mucous membranes, synthesis of rhodopsin; cell growth and cell differentiation; bone development; and immune function; Antioxidant	Regulator of bone mineral metabolism (increase in Ca ²⁺ absorption and promotes bone mineralization), blood calcium homeostasis, cell differentiation, proliferation, and growth; a role in muscle and mitochondrial health, and immune function	Antioxidant, slows blood clotting, linked to the immune system	Assists in blood clotting factor reactions, bone mineralization
Deficiency	Night blindness, increase in infections, impaired growth and wound healing	Weak bones (osteomalacia in adults), poor muscle function, increased risk of infections	Hemolysis, anemia	Bleeding, hemorrhage
Excess or Toxicity	Nausea, headache, fatigue, liver damage, joint pains, peeling of skin, abnormal fetal development in pregnancy	Nausea, loss of appetite, irritability, joint pains, calcification of soft tissues (the kidneys)	Headache, fatigue, diarrhea	Thrombosis, vomiting

Calcium (Ca²⁺). Adapted from Gropper & Smith (2022) and Jeukendrup & Gleeson (2018).